

Enterprise AI Course: Cloud-Based Full-Stack DevSecOps with Retrieval-Augmented Generation

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I. MOTIVATION & PROBLEM STATEMENT

Enterprise AI, a convergence of advanced technologies such as Natural Language Processing (NLP), Machine Learning (ML), Deep Learning (DL), and computer vision, has the potential to revolutionize business operations in large companies. The release of OpenAI's Generative Pre-trained Transformer 3 (GPT-3) on May 28, 2020, sparked a surge of interest in applying Large Language Models (LLMs) to organizations. Meta's innovative proposal of Retrieval-Augmented Generation (RAG), which seamlessly integrates external data into the prompt generation in an LLM (Lewis et al., 2020), further fueled this interest. However, the implementation of enterprise AI projects in production environments is a formidable task, one that can be eased with the use of LLMs and RAG (Diaz-De-Arcaya et al., 2023). This challenge necessitates the application of software engineering principles to the enterprise AI lifecycle. In response, our research proposes an Enterprise AI course design that integrates Cloud-Based Full-Stack DevSecOps with RAG, offering a comprehensive solution to these challenges and instilling confidence in professionals to tackle them effectively.

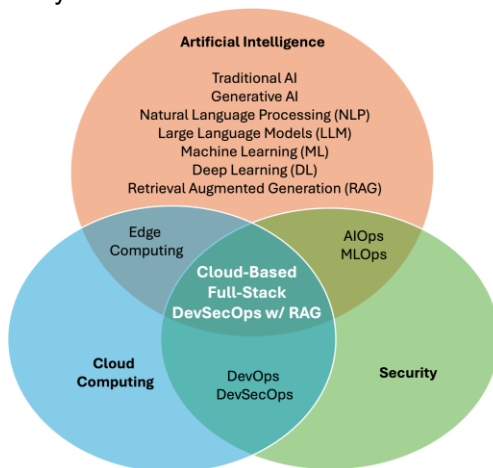


Figure 1: Cloud Computing, Security, & AI Venn Diagram

II. APPROACH

The proposed approach in Figure 1 has the following components: 1) Understand cloud and edge infrastructure and leverage to scale AI applications. 2) Introduce streamlined development and emphasize continuous integration/continuous deployment pipelines. 3) Incorporate security best practices in DevOps and ensure security is integral in the Software Development Lifecycle. 4) Explore Generative AI (GenAI) models and their use in business contexts. 5) Provide an in-depth understanding of ML and DL and apply them in various enterprise use cases. 6) Discuss principles of ML Operations (MLOps) and AI Operations (AIOps) to manage lifecycle ML and AI models and implement best practices for deploying, monitoring, and maintaining AI models in production. 7) Explore NLP techniques and implement NLP models to enhance business operations. 8) Study the architecture and applications of LLMs and describe the capabilities and limitations of LLMs in business contexts. 9) Examine the RAG framework and its integration with LLMs and implement RAG to improve the accuracy and relevance of AI-generated outputs and decrease hallucinations.

III. CONCLUSION

The Enterprise AI course is designed to cover essential components for a successful AI deployment in the enterprise by integrating DevOps, DevSecOps, GenAI, ML, DL, MLOps, AIOps, NLP, LLM, and RAG.

REFERENCES

- Diaz-De-Arcaya, J., Torre-Bastida, A. I., Zarate, G., Minon, R., & Almeida, A. (2023). *A joint study of the challenges, opportunities, and roadmap of MLOps and AIOps: A systematic survey*. *ACM Computing Surveys*, 56(4), 1-30.
- Lewis, P. et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. *Part of Advances in Neural Information Processing Systems 33* (NeurIPS 2020).